Lecture 4

Typing, Trees and Outbreaks

Day 2 · 10:45–11:30 · 45 min

LO5: type isolates and interpret a phylogeny to support or rule out an outbreak

“We’ve had 5 *Klebsiella* SSIs on Ward B in 6 weeks. Is this an outbreak?”

What is an outbreak?

- An outbreak means **transmission**: the same strain passing between patients
- Directly, or through staff, equipment or the environment

KEY MESSAGE

Same species ≠ same strain.
Same AST pattern ≠ same strain.

Typing: naming bacteria at different resolutions

Method	Like a…	Used for
Species	country	Clinical ID
MLST (7 genes) → ST	city	Is it a known high-risk clone?
cgMLST (1,000–3,000 genes)	street	Clusters within an ST
SNPs (whole genome)	house number	Outbreaks and transmission

MLST in one slide

- Seven “housekeeping” genes; each version gets a number
- The combination of 7 numbers = the **Sequence Type (ST)**
- Known high-risk clones: *K. pneumoniae* ST258, ST307 · MRSA ST8, ST22
- Tools do it automatically (Lab 4, Lab 5)

KEY MESSAGE

Same ST = possibly related. Different ST = not a recent outbreak.

SNP distances

- Align genomes to a reference
- Count the positions where they differ
- **Few SNPs** → recent common ancestor → possible transmission
- **Many SNPs** → not recent transmission

ROUGH GUIDE (VARIES BY STUDY)

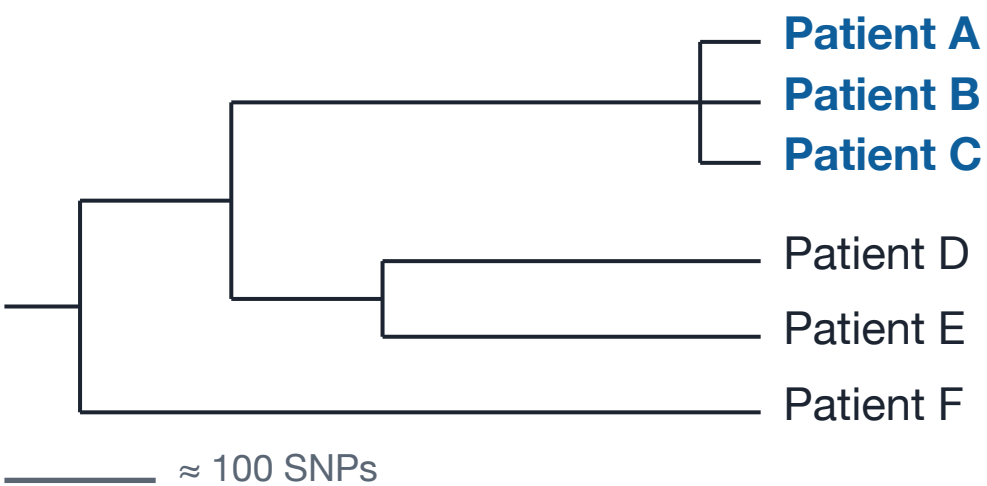
S. aureus: **≤ 15–30 SNPs**

K. pneumoniae, *E. coli*: **≤ 10–25 SNPs**

KEY MESSAGE

Thresholds are guides, not laws. Always combine them with epidemiology.

Reading a phylogenetic tree



- **Tips** = isolates
- **Branch length** = genetic change
- Tips on **short branches** close together are closely related (A, B, C)

COMMON MISTAKES

- Reading top-to-bottom order as meaningful
- Ignoring the **scale bar**

Genomics + epidemiology: time, place, person

A genomic cluster is an outbreak only if the **epidemiology fits**: same ward or theatre at overlapping times? Same surgeon, equipment or staff?

Confirm
Same strain, few SNPs, overlapping stays

Refute
Different STs: no need to close the ward

Find hidden links
Cases on different wards linked by shared equipment or staff

Strain outbreak vs plasmid outbreak

STRAIN OUTBREAK

The **same clone** in many patients

Tree: one tight cluster

PLASMID OUTBREAK

Different strains or species, but the **same resistance plasmid**

Needs Nanopore or hybrid data to confirm

KEY MESSAGE

Tree says “unrelated” but the resistance is unusual and identical? Think plasmid.

Vote: outbreak or not? (cards A / B / C)

1. **4 MRSA**, all ST22, 2–5 SNPs apart, same ward over 3 weeks
2. **4 *Klebsiella***: ST15, ST307, ST14, ST101
3. **3 *E. coli***, all ST131, 80–150 SNPs apart, 3 different hospitals

Vote: answers

1. **Likely outbreak**
2. **Not a single-strain outbreak.** Check for a shared plasmid?
3. **A common circulating clone**, not recent transmission

From result to action

Finding	Possible IPC action
Confirmed cluster	Enhanced cleaning, cohorting, contact screening, theatre audit, staff screening if indicated
Refuted cluster	Avoid unnecessary ward closure; look at other causes (e.g. prophylaxis timing)
High-risk clone or new carbapenemase	Notify the national reference lab / public-health authority

> Communicate in **plain language** to clinicians and managers. You'll practise this in Lab 6.

Tools for the next lab

PATHOGENWATCH: RESULTS

Upload assemblies → species, ST, AMR, clustering, tree

PATHOGENWATCH: CONTEXT

Add a metadata table → tree + timeline + metadata in one interactive view

Free, used by public-health labs, and needs **no coding**.